QUANTUM IAS & PCS

TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

Environment and Ecology

PYQ-Based Revision Pointers

Chapters 1 to 16

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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How to Use This Book

Every pointer is a real question. Each pointer is the answer to a UPPSC question asked between 1990 and 2025, written as a fact you can revise in seconds, with the related fact you need for the next variation.

Frequency marks. ■ and a red count (e.g. 4×) mark a fact asked 3 or more times. • and a gold count mark facts asked once or twice. The years follow; "one-liner" means a short starred question in the source with no year given.

Asked Most box. Each chapter opens with the facts UPPSC repeats most. Revise these first.

Diagrams. Flowcharts and diagrams turn sequences (timelines, procedures, processes, river courses) into one picture.

Trap Alerts. The places where the obvious answer is wrong or two facts are commonly swapped.

Last-Minute 10. The ten lines to read on the morning of the exam.

Pairs with the Masterbook. Use this book for revision after the TATTVA PYQ Masterbook, which has the full questions, options and explanations.

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Chapter 1

Environment, Sustainable Development and Ecology

Built from 44 UPPSC questions (2001–2025). Focus: sustainable development and Brundtland, Silent Spring, Environment Protection Act 1986, ecosystem facts, ecological succession.

ASKED MOST · REVISE FIRST

- ■ **Sustainable development**: meet present needs without compromising future generations (Brundtland, **Our Common Future**, 1987) 4× 2012, 2019, 2020, 2025
- ■ **Ocean** = largest and most stable ecosystem 4× 2003, 2013, 2014, 2018
- ■ Ecosystems are **open** systems, not closed 3× 2014, 2015, 2017
- • **Deforestation** disturbs ecological balance 2× 2013, 2014
- ■ **Ecomark** = label for eco-friendly goods 3× 2004, 2012, 2021
- ■ Succession: **nudation, migration, ecesis, reaction, stabilisation**; the sequence = **sere** 3× 2016, 2020, 2024

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

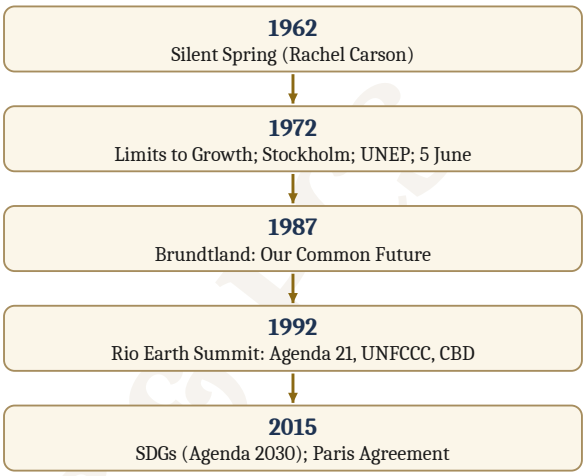
1.1 Environment

- Environment refers to **all** the listed components 1× 2006
- Not a category of environment: **operational environment** 1× 2020
- Environmental degradation means **all** the listed harms 1× 2006
- Not related to environmental protection (key): **poverty alleviation** 1× 2013
- Urbanisation and industrialisation are harmful in **all** listed ways 1× 2002
- Mid-1980s debate on development: **politics and development** was not a theme 1× 2020
- **Plachimada** (Coca-Cola, groundwater): **Kerala** 1× 2009
- **Greenwash**: **false promises** of protecting the environment 1× 2017

1.2 Sustainable Development

- Important for well-being; Brundtland definition (R explains A) 1× 2019
- Report that started the debate (1987): **Our Common Future** 1× 2020
- Sustainable development (true); Agenda 21 signed in 1995 (false, **1992**) 1× 2025
- Order: **Limits to Growth (1972), Brundtland (1987), Montreal Protocol in force (1989), Rio (1992)** 1× 2020
- Inter-generational use of **natural resources** 1× 2012
- Important for sustainable development: **all** listed 1× 2004
- Carrying capacity: waste within absorption capacity (only 2); extraction must stay **below** regeneration 1× 2025
- Sustainable agriculture: use land so its **quality stays intact** 1× 2001

GLOBAL ENVIRONMENT MILESTONES



1.3 Books, Laws, Labels

- **Silent Spring** (1962): **Rachel Carson** 1× 2020
- **The Population Bomb**: **Paul R. Ehrlich** 1× 2023
- **Green Development**: **W.M. Adams** 1× 2011
- **Environment (Protection) Act**: **1986** 1× 2022
- Ecological balance helped by forest, environment, industrial, education policies (all) 1× 2004
- ■ **Ecomark**: **safe goods for the environment** 3× 2004, 2012, 2021

Indian law	Year
Wildlife (Protection) Act	1972
Water Act (set up CPCB)	1974
Forest (Conservation) Act	1980 (Van Adhiniyam 2023)
Air Act	1981
Environment (Protection) Act	1986 (after Bhopal)
Biological Diversity Act	2002
Forest Rights Act	2006
National Green Tribunal Act	2010

1.4 Ecology and Ecosystem

- Ecology: study of **organisms and environment** 1× 2014
- True: ecosystem has biotic and abiotic parts; **autotrophs** make food (1 and 2) 1× 2002
- Not true: an ecosystem is a **closed system** 2× 2014, 2015
- Biosphere is open (true); no new substances can enter (false) 1× 2017
- Order: **population, community, ecosystem, landscape** (then biome, biosphere) 1× 2017

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- True: **producers outnumber primary consumers** 1× 2019
- Components not interdependent (false); humans affect environment (true) 1× 2021
- Largest ecosystem: **ocean**; most stable: **marine** 4× 2003, 2014, 2013, 2018
- Artificial ecosystem: **rice field** 1× 2016
- Cycling of elements: **biogeochemical cycles** 2× 2020, 2021

1.5 Ecological Balance

- Does not help balance: **deforestation**; disturbs it 2× 2013, 2014
- Not related to balance: **industrial management** 1× 2005
- Vegetation is the index of climate; water-loving plants in moist climate (R explains A) 1× 2009

1.6 Succession and Niche

- Phases: **nudation, migration, ecesis, reaction, stabilisation**; from migration if nudation missing 2× 2016, 2024
- Whole sequence of communities: **sere** 1× 2020
- "Ecological transition": **Bennett** 1× 2020
- Niche: tolerance range, resources, role; each species distinct (both) 1× 2022
- Not true: species become **fewer** in a niche when resources are enough 1× 2024

TRAP ALERT

- **Agenda 21** = **1992**, not 1995.
- Ecosystems are **open** for energy.
- **Sere** = sequence; succession = the process.

UP LINK

UP links: the **Environment Protection Act 1986** is enforced in UP by the **UP Pollution Control Board** (Lucknow); UP's **Ganga** and Yamuna cases at the NGT test these laws.

LAST-MINUTE 10

1. Brundtland **1987**: Our Common Future
2. Rio **1992**: Agenda 21; SDGs **2015**
3. Silent Spring = **Rachel Carson**; Population Bomb = **Ehrlich**
4. EPA **1986**; Ecomark = eco-friendly goods
5. Greenwash = false green claims; Plachimada = Kerala
6. Ocean = largest, most stable ecosystem
7. Ecosystem = open system; rice field = artificial
8. Biogeochemical cycles = element cycling
9. Succession: nudation → migration → ecesis → reaction → stabilisation
10. Sere = whole sequence

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Chapter 2

Food Chain, Energy Flow and Ecosystem Concepts

Built from 32 UPPSC questions (2001–2025). Focus: trophic levels, producers, 10% law, energy flow, ecological pyramids, productivity, ecotone and footprint.

ASKED MOST · REVISE FIRST

- **Solar energy** = driving force of ecosystems; energy flow is **one-way** 4× 2015, 2015, 2016, 2019
- Energy **decreases** at each step; **10% law** by **Lindeman** 4× 2017, 2019, 2023, 2025
- Producers = **green plants**; **phytoplankton** in the sea 3× 2005, 2013, 2017
- **Eucalyptus** is not eco-friendly 3× 2005, 2010, 2011

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 Food Chains

- Trophic levels: **organisms linked in a food chain** 1× 2023
- Order: **grass, grasshopper, frog, snake** 1× 2010
- Primary consumers: **ants and deer** 1× 2006
- Man: **primary and secondary** consumer 1× 2016
- Producers: **green plants** 1× 2013
- Main marine producers: **phytoplankton** 1× 2005
- Link between biotic and abiotic: **producers** 1× 2017
- Primary productivity = autotrophs; secondary = heterotrophs (both) 1× 2017
- Correct: **green plants-ecosystem** (producers) 1× 2015

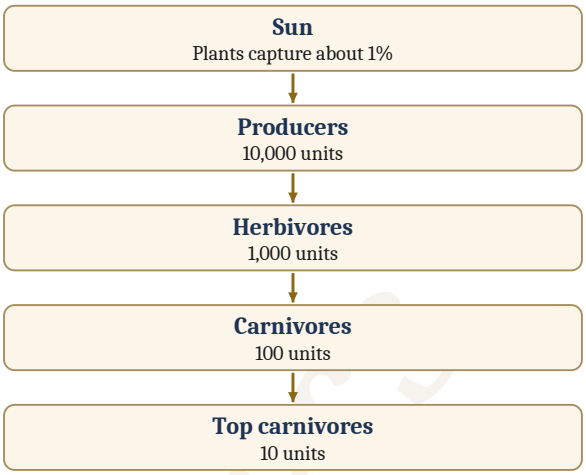
2.2 Aquatic Ecosystems

- **Lotic**: **rivers** (flowing water) 1× 2023
- Phytoplankton thrive in the **limnetic** (open-water) zone (R explains A) 1× 2022

2.3 Energy Flow

- Primary energy source: **solar**; driving force 3× 2015, 2016, 2019
- Energy flow is **unidirectional** 1× 2015
- Energy **decreases** up the levels 2× 2017, 2019
- **Ten Percent Rule**: **Lindeman** 1× 2023
- Plants capture about **1%** of sunlight; about **10%** passes up (both) 1× 2025
- Highest level (key): **omnivores** (top carnivores in most texts) 1× 2002

10% LAW OF ENERGY



2.4 Pyramids and Productivity

- Inverted biomass pyramid: **pond** 1× 2010
- Equatorial forest densest biomass; rain and heat all year (R explains A) 1× 2023
- Least biomass production: **deep sea** 1× 2002
- Decreasing productivity: **mangroves, grasslands, lakes, oceans** 1× 2021

Pyramid	Inverted?
Numbers	Yes in a tree ecosystem
Biomass	Yes in pond and ocean
Energy	Never

2.5 Terms and Thinkers

- Bionomics: synonymous with ecology; values natural systems (2 and 3) 1× 2001
- Land needed to sustain a person: **ecological footprint**; measured in **global hectares** 2× 2012, 2016
- Zone between two communities: **ecotone** 1× 2012
- "Deep ecology": **Arne Naess** 1× 2014
- Monoclimax theory: **F.E. Clements** 1× 2021
- Niche concept: **Grinnell** 1× 2005
- Not eco-friendly: **eucalyptus** 3× 2005, 2010, 2011
- "Ecology is permanent economy": **Chipko**; Chipko = **plant (tree) conservation** 2× 2007, 2015

Term	Thinker
Ecology	Haeckel (1866)
Ecosystem	Tansley (1935)
Niche	Grinnell
10% law	Lindeman
Monoclimax	Clements
Deep ecology	Arne Naess

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TRAP ALERT

- **Energy flows** one way; **nutrients cycle**.
- Energy pyramid **never** inverts; pond biomass pyramid does.
- **Lotic** = flowing; **lentic** = still.

UP LINK

UP link: **Chipko** began in the Garhwal hills (then in UP, now Uttarakhand) with Sunderlal Bahuguna and Chandi Prasad Bhatt; slogan "Ecology is permanent economy".

LAST-MINUTE 10

1. Sun = energy source; flow is one-way
2. 10% law = **Lindeman**; plants catch ~1%
3. Producers: green plants; phytoplankton at sea
4. Grass → grasshopper → frog → snake
5. Man = primary + secondary consumer
6. Pond = inverted biomass pyramid
7. Mangroves most productive; deep sea least
8. Ecotone = transition zone; footprint = global hectares
9. Deep ecology = **Arne Naess**; niche = **Grinnell**
10. Eucalyptus not eco-friendly; Chipko = permanent economy

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Chapter 3

Biodiversity, Its Loss and Conservation

Built from 29 UPPSC questions (2001–2023). Focus: meaning, tropical rainforests, Western Ghats and hotspots, habitat destruction, in-situ and ex-situ conservation.

ASKED MOST · REVISE FIRST

- ■ Biggest cause of biodiversity loss: **habitat destruction** 5× 2002, 2015, 2016, 2017, 2023
- ■ Maximum biodiversity: **tropical rainforests** 4× 2012, 2014, 2016, 2018
- ■ India's richest region: **Western Ghats**; Vindhyas are not a hotspot 4× 2014, 2015, 2016, 2019
- • Coral bleaching: **rising sea temperature** (global warming) 2× 2012, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 Meaning and Distribution

- • Biodiversity: **range of species in an environment**; diverse plants and animals of an area 2× 2014, 2014
- • Term "biodiversity" first used by **Walter G. Rosen** (1985) 1× 2013
- • Most significant aspect: **maintaining the ecosystem** 1× 2015
- • UN decade 2011-20: **Biodiversity** 1× 2012
- ■ Most biodiversity: **tropical rainforest** 4× 2012, 2014, 2016, 2018
- • Not true: diversity **increases towards the poles** (it decreases) 1× 2019
- • Most butterflies in the tropics; they can't bear cold (R explains A) 1× 2019

3.2 India and Hotspots

- • Richest biodiversity (of options): **Silent Valley**; Silent Valley is in **Kerala** 2× 2003, 2008
- • **Valley of Flowers**: **Uttarakhand** 1× 2002
- • Hotspot region (of options): **Andaman and Nicobar** (Sundaland) 1× 2009
- ■ Most biodiverse area: **Western Ghats** 3× 2014, 2015, 2019
- • Not a hotspot: **Vindhyas** 1× 2016

India's hotspots	Area
Himalaya	Indian Himalaya
Indo-Burma	North-east India
Western Ghats and Sri Lanka	Western Ghats
Sundaland	Nicobar Islands

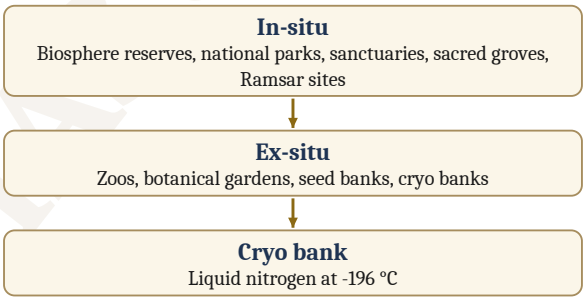
3.3 Causes of Loss

- • Destruction due to **all** listed causes 1× 2002
- • Maximum danger: **habitat and vegetation destruction** 1× 2017
- ■ Main reason: **natural habitat destruction** 3× 2015, 2016, 2023
- • Bird decline: **habitat loss and pesticide use** (3 and 4) 1× 2001
- • Not adding biodiversity: **fewer trophic levels** 1× 2015
- • Coral bleaching: **rise in sea temperature**; **global warming** 2× 2012, 2015

3.4 Conservation

- • Important strategy: **biosphere reserves** 1× 2013
- • Biosphere reserves conserve **genetic variation** 1× 2004
- • Not in-situ: **botanical garden** 1× 2018
- • Cryo-bank gas: **liquid nitrogen** (-196 °C) 1× 2009

CONSERVATION METHODS



TRAP ALERT

- • **Habitat destruction** beats pollution and hunting as the main cause.
- • **Botanical gardens, zoos** = ex-situ; **sacred groves** = in-situ.
- • **Vindhyas** are not a hotspot.

UP LINK

UP's biodiversity sits mostly in the **Terai** (Dudhwa, Katarniaghat, Pilibhit) and the **Himalayan foothill** belt it shares with Uttarakhand; UP has no hotspot area of its own.

LAST-MINUTE 10

1. Term "biodiversity": **Walter G. Rosen**
2. UN Biodiversity Decade **2011-20**
3. Tropical rainforests richest; diversity falls toward poles
4. Western Ghats richest in India
5. Hotspots: Himalaya, Indo-Burma, W. Ghats, Sundaland
6. Silent Valley = Kerala; Valley of Flowers = Uttarakhand
7. Main loss cause: **habitat destruction**
8. Coral bleaching = warmer seas
9. Botanical garden = ex-situ
10. Cryo bank = liquid nitrogen

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Chapter 4

Endangered Species, Ramsar Convention and Biodiversity Miscellany

Built from 21 UPPSC questions (2002–2021). Focus: Red Data Book, vultures and diclofenac, Ramsar wetlands, CBD Nagoya, Biodiversity Act and NBA.

ASKED MOST · REVISE FIRST

- ■ Vultures nearly wiped out by **diclofenac** given to cattle 3× 2008, 2015, 2018
- ■ **Ramsar Convention** = **wetlands**; World Wetlands Day **2 February** 3× 2008, 2008, 2016
- • **Red Data Book** lists endangered species 2× 2002, 2013
- • **Biological Diversity Act**: **December 2002**; **NBA** 2003, **Chennai** 2× 2010, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

4.1 Endangered Species

- Endangered species listed in the **Red Data Book**; species facing extinction 2× 2013, 2002
- Highly endangered Indian bird: **Great Indian Bustard** 1× 2005
- Wrong: **desert cat-endangered** 1× 2014
- Extinct if unseen for **50 years** (old rule; IUCN now has no fixed period) 1× 2005
- ■ Vulture killer: **diclofenac**; painkillers to cattle 3× 2008, 2018, 2015

IUCN category	Meaning
Critically Endangered	Extremely high risk
Endangered	Very high risk
Vulnerable	High risk
Near Threatened	Close to threatened
Least Concern	Widespread

4.2 Ramsar and Wetlands

- Ramsar Convention (1971): **wetlands** 2× 2008, 2016
- Wetlands Day: **2 February** 1× 2008
- Ramsar sites: **Chilika, Keoladeo, Wular, Loktak** (all) 1× 2013
- Ramsar site of the options: **Sundarbans** 1× 2015
- Ramsar site in a State: **Point Calimere** (Tamil Nadu) 1× 2021
- Wetlands useful for **all** listed services 1× 2012

4.3 Laws and Institutions

- CBD COP10: **Nagoya** (2010) 1× 2010
- Biodiversity Act passed: **December 2002** 1× 2010
- NBA set up: **2003, Chennai** 1× 2014
- Centre for Biodiversity Policy and Law (with Norway): **Chennai** 1× 2017

CBD TIMELINE



4.4 Miscellany

- Not a cause of extinction: **broad niche** 1× 2014
- **Sea buckthorn**: **Ladakh** 1× 2004
- Largest ecosystem: **biosphere** 1× 2017
- "Tomorrow's Biodiversity": **Vandana Shiva** 2× 2004, 2005

TRAP ALERT

- Wetlands Day **2 Feb**; Environment Day **5 June**.
- **CBD COP10 = Nagoya**; climate COP15 = Copenhagen.
- A **broad** niche protects; a narrow one endangers.

UP LINK

UP's Ramsar sites include **Nawabganj, Parvati Agra, Saman, Samaspur, Sandi, Sarsai Nawar, Sur Sarovar, Upper Ganga (Brijghat-Narora)**, **Bakhira** and **Haiderpur** wetlands.

LAST-MINUTE 10

1. Red Data Book = endangered species
2. Diclofenac killed vultures
3. Great Indian Bustard = highly endangered
4. Ramsar = wetlands (1971); Wetlands Day **2 Feb**
5. Chilika, Keoladeo, Wular, Loktak = Ramsar sites
6. Point Calimere = Tamil Nadu
7. CBD COP10 = **Nagoya**
8. Biodiversity Act **2002**; NBA **2003 Chennai**
9. Sea buckthorn = Ladakh
10. Tomorrow's Biodiversity = **Vandana Shiva**

Chapter 5

Greenhouse Effect and Climate Change

Built from 40 UPPSC questions (2001–2024). Focus: which gases are greenhouse gases, CO2 and methane, effects of warming, emitters, net zero, Milankovitch.

ASKED MOST · REVISE FIRST

- **Not greenhouse gases**: **N2, O2, argon, H2, SO2, NO2, propane** 11× 2007, 2012, 2012, 2016, 2016, 2016, 2017, 2017, 2019, 2020, 2022
- **CO2** = main cause of global warming 6× 2004, 2004, 2006, 2008, 2013, 2015
- **Methane** from **paddy fields** and cattle 3× 2017, 2018, 2019
- **China** = largest CO2 emitter 3× 2010, 2015, 2018
- Warming effects: sea-level rise, melting glaciers, disease, coral bleaching, early mango flowering 4× 2001, 2002, 2003, 2020

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Greenhouse Effect

- Concept of greenhouse gases: **Joseph Fourier** 1× 2011
- Greenhouse effect: **rise in global temperature** 1× 2018
- Without it Earth would average **-18 °C** 1× 2020
- Enhanced by **CO2**; main gases **CO2 and methane** 3× 2015, 2004, 2013
- Largest natural share: **water vapour** 1× 2011
- Not responsible: **argon** 2× 2016, 2022
- Not directly warming: **oxides of sulphur and nitrogen** 1× 2013
- Not a natural GHG (of options): **nitrogen oxide** 2× 2019, 2020
- Not GHG: **SO2**; **oxygen**; **NO2**; **propane**; **nitrogen**; **hydrogen** 7× 2007, 2016, 2017, 2016, 2017, 2012, 2012
- Both harmful and useful: **CO2** per key (ozone in most texts) 1× 2013

Gas	Main source	GWP (100 yr)
CO2	Fossil fuels, deforestation	1
Methane	Paddy, cattle, wetlands, landfills	~28
Nitrous oxide (N2O)	Fertilisers	~265
CFCs, HFCs, SF6	Industry	Thousands

5.2 CO2 and Methane

- Global warming from **CO2**; worry grows with CO2 5× 2004, 2013, 2004, 2006, 2008
- Natural optimum CO2: **0.03%** (now about 0.042%) 1× 2017
- Does not add CO2: **photosynthesis** 1× 2012
- Methane from **both** listed sources 1× 2018
- Paddy-field gas: **methane** 2× 2017, 2019

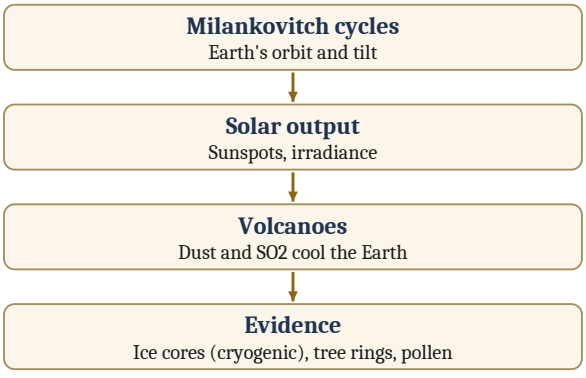
5.3 Causes and Effects

- Climate change from **all** listed causes 1× 2013
- Primary causes: **fossil fuels, cars, deforestation**; not solar flares (1, 2, 4) 1× 2017
- Effect of greenhouse effect: unclear question (*) 1× 2017
- Effects: **melting glaciers, early mango flowering, health harm** (2, 3, 4) 3× 2001, 2002, 2003
- Effects: sea-level rise, glaciers, disease, coral bleaching (all) 1× 2020
- More frequent: **cyclones, storms, hurricanes** 1× 2018
- Andaman coral bleaching: **2 °C warmer Indian Ocean** 1× 2024
- Warming last century: about **0.9 °C** 1× 2012
- **Urban heat island**: cities hotter than nearby rural areas 1× 2023
- Climate change raises social tension via food insecurity (R explains A) 1× 2017

5.4 Emitters and Net Zero

- Largest CO2 emitter: **China** 3× 2010, 2015, 2018
- Carbon-negative country: **Bhutan** 1× 2015
- Net zero = emissions **balanced by removals**, not zero (only 2) 1× 2021
- IPCC: **deep, rapid, sustained** cuts essential (only 2) 1× 2024

NATURAL CAUSES OF CLIMATE CHANGE



5.5 Natural Causes

- Not astronomical theory: **solar irradiance** 1× 2015
- Axis shift as climate factor: **Milutin Milankovitch** 1× 2015
- Cryogenic indicator: **ice core** 1× 2015

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TRAP ALERT

- **N2O** is a GHG; **NO2** is not.
- N2, O2, Ar make up air but are **not** GHGs.
- **Net zero** is not zero emissions.

UP LINK

UP links: UP's large **paddy and livestock** sector makes it a big methane source; **Lucknow and Kanpur** show urban heat-island effects in heatwaves.

LAST-MINUTE 10

1. Greenhouse idea: **Fourier**; without it **-18 °C**
2. CO2 = main human driver; water vapour = largest natural share
3. Methane = paddy, cattle
4. Not GHG: N2, O2, Ar, H2, SO2, NO2, propane
5. CO2 natural level **0.03%**
6. Warming ~0.9 °C last century
7. China = largest emitter; Bhutan = carbon negative
8. Net zero = balance, not zero
9. Milankovitch = orbit; ice core = cryogenic proof
10. Urban heat island = hotter cities

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Chapter 6

Climate Change Conferences, Treaties and Action Plans

Built from 25 UPPSC questions (2001–2025). Focus: Agenda 21, UNFCCC, Kyoto and carbon credits, Paris COP21, GCF and GEF, NAPCC missions.

ASKED MOST · REVISE FIRST

- ■ **Agenda 21** (Rio 1992) = action plan for **sustainable development** 3× 2013, 2014, 2023
- ■ **Kyoto Protocol** = **climate change**; source of **carbon credits** 4× 2015, 2017, 2019, 2021
- • **COP21 = Paris** (2015) 2× 2015, 2018
- ■ **NAPCC** released **2008**; 8 missions, **no nuclear** mission 5× 2010, 2016, 2016, 2018, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Earth Summits and UNFCCC

- Earth Summit + 5: **1997** 1× 2016
- ■ Agenda 21: action plan for the **21st-century environment**; **sustainable development** 3× 2013, 2014, 2023
- Wrong: **carbon trading-Montreal Protocol** (it is Kyoto) 1× 2015
- UNFCCC in force: **21 March 1994** 1× 2012
- Paris conference: **COP21** 2× 2015, 2018

6.2 Protocols and Funds

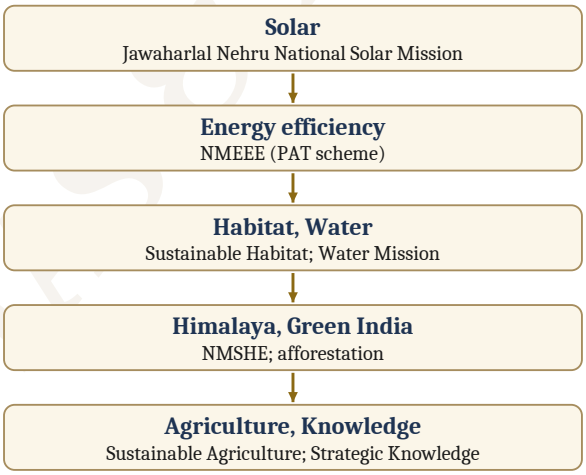
- Wrong: **Kyoto-save water**; Kyoto = **climate change** 2× 2019, 2017
- Incorrect: GCF set up at **Durban** (it was **Cancun**, 2010) 1× 2013
- GEF set up **1991** (true); only rich countries donate (false) (only 1) 1× 2025
- Not true of **CDM**: it bars rich countries investing in poor ones 1× 2001
- Carbon credits come from the **Kyoto Protocol** 1× 2021
- Not correct: World Bank's 2011 ERPA was signed with the **Government of India** 1× 2010
- Carbon tax from 2019: **Singapore** 1× 2017
- UN Climate Action Summit 2019: **New York** 1× 2019

COP	Place	Outcome
COP3 1997	Kyoto	Kyoto Protocol
COP15 2009	Copenhagen	$100 bn pledge
COP16 2010	Cancun	Green Climate Fund
COP21 2015	Paris	Paris Agreement
COP26 2021	Glasgow	India net zero 2070
COP27 2022	Sharm el-Sheikh	Loss and Damage fund
COP28 2023	Dubai	Global Stocktake
COP29 2024	Baku	$300 bn finance goal

6.3 NAPCC

- ■ First NAPCC: **2008** 3× 2016, 2018, 2021
- • Not among the 8 missions: **nuclear power** 2× 2010, 2016
- • Mission start years: sources differ (*) 1× 2016

NAPCC'S 8 MISSIONS (2008)



6.4 Miscellany

- **Great Barrier Reef**: NE Australia (true); over 2,500 km (false per key) (only 1) 1× 2023
- **Deep Carbon Observatory**: carbon research and deep microbes (both) 1× 2019
- Cloud brightening uses **sea water** 1× 2009
- Ozone layer blocks UV; depletion linked to CFCs (both) 1× 2025
- No global atmosphere watch station then: **India** 1× 2012
- **SAFAR** inaugurated (2018) by **Dr. Harsh Vardhan** 1× 2018

TRAP ALERT

- **GCF = Cancun**, not Durban; **Kyoto = 1997**.
- **Montreal** = ozone; **Kyoto** = carbon trading.
- Nuclear power is **not** an NAPCC mission.

UP LINK

UP has its own **State Action Plan on Climate Change** under the NAPCC; it targets solar parks in **Bundelkhand** and Green India plantation drives.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Agenda 21 = Rio 1992; sustainable development
2. Earth Summit + 5 = **1997**
3. UNFCCC in force **21 March 1994**
4. Kyoto **1997** = carbon credits, CDM
5. Paris = **COP21** (2015)
6. GCF = **Cancun 2010**; GEF = 1991
7. Singapore = carbon tax 2019
8. NAPCC **2008**; 8 missions; no nuclear
9. Cloud brightening = sea water spray
10. SAFAR = air-quality forecasting (2018)

QUANTUM IAS & PCS

Chapter 7

The Ozone Layer

Built from 19 UPPSC questions (1996–2023). Focus: stratospheric ozone, UV protection, CFCs and other depleters, the ozone hole, Ozone Day.

ASKED MOST · REVISE FIRST

- Ozone layer: **stratosphere** 6× 2008, 2010, 2012, 2018, 2018, 2023
- Ozone absorbs **ultraviolet** rays; protects life 5× 2004, 2006, 2010, 2013, 2014
- **CFCs** deplete ozone 3× 2009, 2014, 2016
- Ozone Day: **16 September** (Montreal Protocol, 1987) 4× 1996, 2014, 2015, 2022

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

7.1 Ozone and Location

- UV light forms **O3** 1× 2018
- Ozone layer: **stratosphere** 5× 2008, 2010, 2012, 2018, 2023
- Wrong: **ozone layer-troposphere** 1× 2018
- Natural regulator of stratospheric ozone (per key): **nitrogen dioxide** (NOx) 1× 2016

7.2 What Ozone Does

- Absorbs **ultraviolet** radiation; protects from **UV rays** 2× 2013, 2014
- Shields life from UV 2× 2004, 2006
- Ozone essential for life; blocks high-energy radiation (R explains A) 1× 2010
- UV causes **skin cancer** 1× 2014

7.3 Depleting Substances

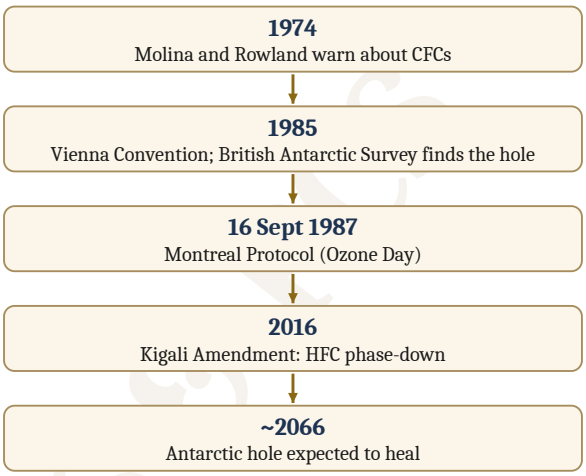
- Main depleter: **CFCs** 3× 2009, 2014, 2016
- Not true: CFCs are **not** greenhouse gases (they are) 1× 2004
- Depleters: **CFCs, halons, carbon tetrachloride** (all) 1× 2012
- Not a depleter (key): **methyl chloroform** (actually it is one) 1× 2002
- Wrong: **CFC-automobile** (CFCs from refrigerants, sprays) 1× 2017
- Does not deplete ozone: **carbon monoxide** 1× 2008

Substance	Use
CFCs	Refrigerators, ACs, aerosols
Halons	Fire extinguishers
Carbon tetrachloride	Solvent
Methyl bromide	Fumigant
HFCs	No ozone harm; strong GHGs (Kigali 2016)

7.4 The Ozone Hole

- Antarctic ozone hole found by a **British team** (Farman, 1985) 1× 2013
- Ozone "halo" over Tibet (2005): **G.W. Kent Moore** 1× 2013
- Ozone Day: **16 September** 4× 1996, 2014, 2015, 2022

SAVING THE OZONE LAYER



TRAP ALERT

- Ozone absorbs **UV**, not infrared.
- Ground-level ozone is a **pollutant**; stratospheric ozone protects.
- CFCs **are** greenhouse gases too.

UP LINK

UP's refrigeration and foam industries (Noida, Ghaziabad) switched from CFCs under India's **Montreal Protocol** phase-out; HCFCs end by 2030.

LAST-MINUTE 10

1. Ozone layer = **stratosphere** (15-35 km)
2. Absorbs **UV-B**; UV causes skin cancer, cataract
3. CFCs, halons, CCl4 deplete ozone
4. CFCs also greenhouse gases
5. CO does not deplete ozone
6. Ozone hole: British team, 1985
7. Tibet ozone halo: Kent Moore
8. Montreal Protocol **1987**
9. Ozone Day **16 September**
10. Kigali **2016** = HFCs

QUANTUM IAS & PCS

Chapter 8

Forests and Their Types

Built from 17 UPPSC questions (2002–2023). Focus: forest types and their regions, tropical rainforest records, mangroves, forests as a resource.

ASKED MOST · REVISE FIRST

- Most plant diversity and biomass: **tropical evergreen / rain forest** 2× 2017, 2017
- Mangroves: **Sundarbans**; **Bhitarkanika** is in **Odisha** 3× 2004, 2016, 2016
- **Western Ghats** = evergreen forests 1× 2015
- Forest types: **tropical-Silent Valley, coniferous-Himachal, mangrove-Sundarbans, deciduous/thorn-Rajasthan** 1× 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

8.1 Types and Productivity

- Maximum plant diversity: **tropical evergreen forests** 1× 2017
- Maximum plant biomass: **tropical rainforest** 1× 2017
- Lowest net primary productivity: **temperate grasslands** 1× 2015
- Tropical deciduous forests most extensive in **Asia** 1× 2014
- Evergreen forest: **Western Ghats** 1× 2015
- Wrong: **Eastern Deccan Plateau-moist forests** (dry deciduous) 1× 2013
- **Tropical forest-Silent Valley, coniferous-Himachal Pradesh, mangroves-Sundarbans, deciduous-Rajasthan** 1× 2017

Forest type	Rainfall	Example region
Tropical evergreen	Over 200 cm	Western Ghats, NE, A&N
Moist deciduous	100-200 cm	Sal, teak belt
Dry deciduous	70-100 cm	Central India, UP
Thorn	Under 70 cm	Rajasthan, Gujarat
Montane coniferous	Altitude	Himalaya (deodar, pine)
Mangrove	Tidal deltas	Sundarbans, Bhitarkanika

8.2 Mangroves

- Famous for mangroves: **Sundarbans** 1× 2004
- Protected mangrove region (of options): **Goa** 1× 2016
- Bhitarkanika: not Vansadhara-Subarnarekha delta, not West Bengal (**neither**; it is Brahmani-Baitarani, Odisha) 1× 2016
- Coastal Odisha cyclone-prone; mangrove loss in Mahanadi delta (key: R explains A) 2× 2006, 2010

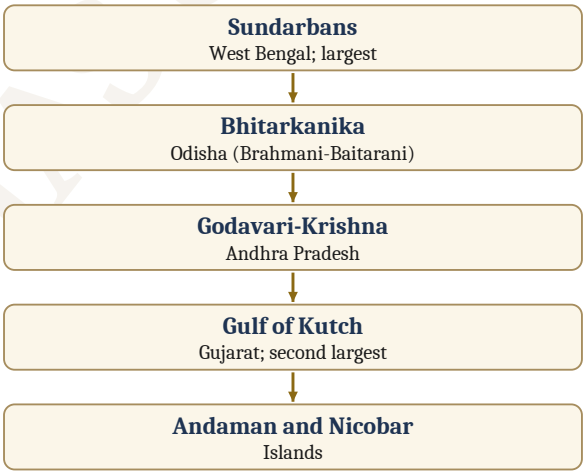
8.3 Forests as a Resource

- Forests renewable; improve the environment (both true, R not the explanation) 2× 2002, 2004
- Hydrological cycle controlled by **forests** (per key) 1× 2023
- Natural resources usable later; development depends on technology (both true, R not the explanation) 1× 2016
- Aluminium a green metal (true); substitutes wood (false) 1× 2009

8.4 Causes of Loss

- Forest loss linked to demographic transition; inverse to population (R explains A) 1× 2016
- Main cause (key): **industrial development** (FAO: farm expansion) 1× 2011

MANGROVE BELTS OF INDIA



TRAP ALERT

- **Bhitarkanika** = Odisha, not West Bengal.
- **Evergreen** = Western Ghats; **coniferous** = Himalaya.
- Lowest NPP of options = **temperate grassland**.

UP LINK

UP's forests are mostly **tropical dry and moist deciduous** (sal in the Terai; teak, tendu in Bundelkhand and Sonbhadra); UP has no mangroves.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Tropical evergreen: most plant diversity and biomass
2. Western Ghats = evergreen
3. Coniferous = Himalaya; thorn = Rajasthan
4. Mangroves: **Sundarbans** largest; Bhitarkanika = Odisha
5. Gulf of Kutch = 2nd mangrove area
6. Asia = most tropical deciduous forest
7. Temperate grassland = lowest NPP
8. Forests control the water cycle
9. Aluminium = green metal
10. Forest loss rises with population

QUANTUM IAS & PCS

Chapter 9

India State of Forest Report (ISFR)

Built from 20 UPPSC questions (1991–2022). Focus: national forest percentages by report year, state rankings by area and percentage, very dense forest, UP.

ASKED MOST · REVISE FIRST

- ■ Highest forest **percentage**: **Mizoram** 5× 2002, 2008, 2010, 2012, 2016
- • Largest forest **area**: **Madhya Pradesh** 1× 2000
- ■ Most **very dense** forest: **Arunachal Pradesh** 3× 2004, 2008, 2011
- • National cover by report: **21.54%** forest (2017); **24.16%** forest + tree (2015) 2× 2016, 2018

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

9.1 National Figures

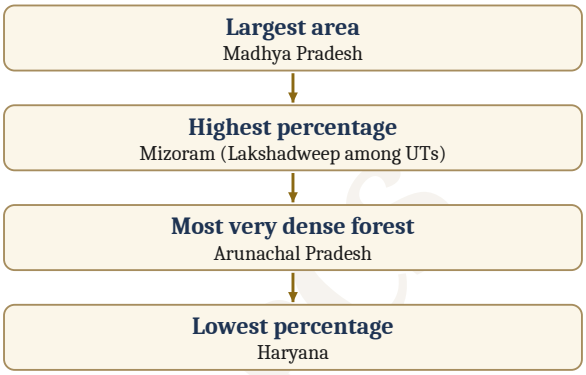
- Forest share (1991 key): **22%** 1× 1991
- ISFR 2017: **21.54%** under forest 1× 2018
- ISFR 2015: **24.16%** forest and tree cover 1× 2016
- Dense forest share: about **12%** 1× 2008

ISFR 2023	Figure
Forest cover	21.76%
Forest + tree cover	25.17%
Mangroves	~4,992 sq km
Target (Forest Policy 1988)	33%

9.2 State Rankings

- ■ Largest forest percentage: **Mizoram**; among States/UTs: **Mizoram** 5× 2002, 2010, 2012, 2016, 2008
- • Most forest of HP, Arunachal, Meghalaya, Sikkim: **Arunachal** 1× 2004
- • Over 75% forest group: none fits now (*) 1× 1995
- • Over 80% forest (then): **Arunachal**; now **Mizoram** 1× 2008
- • Over 33% forest (ISFR 2021): **17** states/UTs 1× 2021
- • Largest forested area: **Madhya Pradesh** 1× 2000
- • Largest very dense forest: **Arunachal Pradesh** 1× 2011
- • Not correct: **Nagaland** is the most forested State 1× 2010
- • Most forest gain (ISFR 2021): **Andhra Pradesh** 1× 2022
- • Second largest mangroves: **Gujarat** 1× 2008
- • Tree cover maximum (key): **East Deccan** 1× 2009
- • Descending cover: **Manipur, Odisha, Maharashtra, Haryana** 1× 2002
- • Forest area order: Arunachal, Chhattisgarh, Odisha, AP (no code, *) 1× 2011
- • Most tribal districts (ISFR 2019 options): **Manipur** 1× 2021

THREE DIFFERENT "FIRSTS"



9.3 Uttar Pradesh

- UP's recorded forest area (2011): about **7%** of its area 1× 2012

TRAP ALERT

- **Area** (MP) vs **percentage** (Mizoram) vs **very dense** (Arunachal).
- "Recorded forest area" (legal) differs from "forest cover" (satellite).
- Answers change by report year; check the ISFR edition.

UP LINK

UP has only about **6%** forest cover in recent ISFRs, far below the 33% goal; the State's mass plantation drives (crores of saplings each July) aim to raise green cover.

LAST-MINUTE 10

1. ISFR by **Forest Survey of India** (Dehradun), every 2 years
2. 2023: forest **21.76%**; forest + tree **25.17%**
3. Target **33%** (two-thirds in hills)
4. Mizoram = highest %; MP = largest area
5. Arunachal = most very dense forest
6. Haryana = lowest %
7. West Bengal, Gujarat = top mangroves
8. 17 States/UTs over 33% (2021)
9. Andhra gained most (2021)
10. UP ~7% recorded forest

Chapter 10

Forest and Wildlife Conservation Efforts

Built from 26 UPPSC questions (1998–2025). Focus: Forest Policy and the 33% goal, Wildlife and Forest Acts, schemes and operations, institutions, wildlife facts.

ASKED MOST · REVISE FIRST

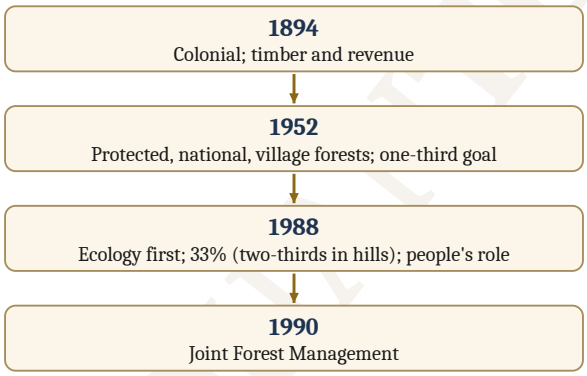
- Forest goal: **33%** (one-third) of area 2× 2002, 2014
- **Wildlife (Protection) Act 1972**; **Forest (Conservation) Act 1980** 4× 2013, 2015, 2017, 2021
- Order: **Wildlife Act (1972), Project Tiger (1973), Project Elephant (1992), Biodiversity Act (2002)** 2× 2019, 2020
- **Forest Research Institute**: **Dehradun** 2× 2008, 2016
- WWF symbol: **giant panda** 3× 2001, 2002, 2003

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

10.1 Forest Policy

- Forest area for ecological balance: **33%**; one-third 2× 2014, 2002
- Not a forest class in the 1952 policy: **national park** 1× 2016
- Policy aims (key): **ecological balance, people's participation** (1 and 4) 1× 2001

FOREST POLICIES



10.2 Laws

- Wildlife Protection Act: **1972** 1× 2015
- Forest (Conservation) Act: **1980** 2× 2013, 2017
- **Forest Conservation-1980, Environment Protection-1986, Air-1981, Water-1974** 1× 2021
- Order: **Wildlife Act, Project Tiger, Project Elephant, Biodiversity Act** 2× 2019, 2020
- Wildlife Act protects only animals (false); Project Elephant only Asian elephants (false): **neither** 1× 2023

10.3 Schemes and Operations

- **MISHTI**: restores mangroves (true); begun 2019 (false, **2023**) 1× 2025
- **Operation Olivia**: Coast Guard guards olive ridleys off Odisha (true); in August (false) 1× 2025
- "Apna Van Apna Dhan": **Himachal Pradesh** 1× 2012
- Plantation town: **Valparai** 1× 2013
- **SAVE** (India-Nepal): **vultures** 1× 2011
- New Forest Inventory (2016) revisit: **5 years** 1× 2017

10.4 Institutions

- **Forest Research Institute**: **Dehradun** 2× 2008, 2016
- **NBPGR**: **New Delhi** 1× 2013
- Rajiv Gandhi Wildlife Conservation Award: **all** listed 1× 1998

Institution	Location
FRI, FSI, Wildlife Institute of India	Dehradun
Forest Management Institute	Bhopal
NBPGR	New Delhi
BSI, ZSI	Kolkata
NBA	Chennai
SACON	Coimbatore

10.5 Wildlife Facts

- India's share of world wildlife: about **6.5-7.5%** (*) 1× 2008
- India's largest fish: **whale shark** 1× 2008
- Not a major extinction cause (key): **fast population growth** 1× 2020
- **Pugmark** technique: **counting wild animals** 1× 2008
- ■ WWF symbol: **giant panda** 3× 2001, 2002, 2003
- Statements 1, 2, 4 correct; eucalyptus is not eco-friendly 1× 2002
- Largest forest share (of options): **Suriname**; **Japan** 2× 2009, 2013

TRAP ALERT

- **MISHTI = 2023**; olive ridleys nest **Jan-March**.
- Wildlife Act also protects **plants**.
- National parks are a Wildlife Act category, not a 1952 forest class.

UP LINK

UP links: **Project Tiger** covers **Dudhwa, Pilibhit, Amangarh** and **Ranipur** tiger reserves; UP runs the giant one-day **Van Mahotsav** plantation drives.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Forest goal **33%**; hills two-thirds
2. Wildlife Act **1972**; Forest Conservation Act **1980**
3. Water **1974**, Air **1981**, EPA **1986**
4. Project Tiger **1973**; Elephant **1992**; Biodiversity Act **2002**
5. MISHTI (2023) = mangroves
6. Operation Olivia = olive ridleys, Odisha
7. SAVE = vultures
8. FRI = **Dehradun**; NBPGR = **New Delhi**
9. Pugmark = animal census
10. WWF symbol = **giant panda**

QUANTUM IAS & PCS

Chapter 11

National Parks

Built from 41 UPPSC questions (1996–2024). Focus: park-state matching, park-animal matching, first parks, marine parks.

ASKED MOST · REVISE FIRST

- **Park-state matching**: **Bandipur-Karnataka, Similipal-Odisha, Pin Valley and Rohla-Himachal, Indravati-Chhattisgarh, Dachigam-J&K, Intanki-Nagaland, Betla-Jharkhand, Guindy-TN** 12× 1998, 2002, 2002, 2002, 2010, 2015, 2016, 2016, 2017, 2018, 2020, 2024
- **Corbett** = first national park (1936); fed by the **Ramganga** 2× 2006, 2017
- Animals: **Kaziranga-rhino, Gir-lion, Sundarbans-tiger, Periyar-elephant, Dachigam-hangul** 4× 2006, 2014, 2018, 2022
- **Dudhwa** (UP) = rhino rehabilitation 2× 2004, 2022
- Marine park: **Gulf of Kutch** (first), **Gulf of Mannar** 3× 2013, 2015, 2016

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

11.1 First Parks

- First national park: **Jim Corbett**; water from **Ramganga** 2× 2017, 2006
- **Yellowstone**: **USA** 3× 2004, 2010, 2011
- Order: **Corbett, Kanha, Kaziranga, Silent Valley** 1× 2016
- Genetic diversity in parks: **in-situ** conservation 1× 2023
- Most national parks: **Madhya Pradesh** 2× 2004, 2008
- Most parks in MP; most sanctuaries in A&N (both) 1× 2021

11.2 Park and State

- **Rajiv Gandhi NP** (Nagarhole): **Karnataka** 1× 2002
- **Keibul Lamjao**: on **Loktak** lake 1× 2022
- Wrong: **Pin Valley-J&K** (Himachal) 2× 2016, 2024
- Wrong: **Similipal-MP** (Odisha) 1× 2018
- Wrong: **Bandipur-Tamil Nadu** (Karnataka) 1× 2010
- True: Similipal in Odisha; Siam = Thailand (1 and 2) 1× 2002
- **Bandhavgarh-MP, Bandipur-Karnataka, Rohla-Himachal, Gir-Gujarat** 1× 2002
- **Uttarakhand-Rajaji, Assam-Manas, Odisha-Similipal, Karnataka-Bandipur** 1× 1998
- **Dachigam-J&K, Papikonda-AP, Sariska-Rajasthan, Bandipur-Karnataka** 1× 2015
- **Assam-Kaziranga, Gujarat-Gir, MP-Kanha, Uttarakhand-Corbett** 1× 2002
- **Indravati**: **Chhattisgarh**; in **Bastar** 2× 2016, 2015
- **Intanki-Nagaland, Betla-Jharkhand, Shiroi-Manipur, Guindy-Tamil Nadu** 1× 2017
- **Indravati-Chhattisgarh, Mollem-Goa, Kalesar-Haryana, Betla-Jharkhand** 1× 2020
- Correct: **Keoladeo-Bharatpur** 1× 2009
- **Kaziranga-Golaghat, Kudremukh-Chikmagalur, Silent Valley-Palghat, Pench-Nagpur** 1× 2013
- **Dhoopgarh** peak: **Satpura NP** 1× 2021

11.3 Parks and Animals

- Wrong: **Ghana (Keoladeo)-lion** (birds) 1× 2009
- **Kaziranga-rhino, Gir-lion, Sundarbans-tiger, Periyar-elephant** 1× 2006
- Wrong: **Periyar-hangul** 1× 2014
- **Dachigam**: **hangul** (Kashmir stag); hangul in **J&K** 2× 2018, 2022
- Rhino rehabilitation: **Dudhwa** 2× 2004, 2022
- Siberian crane habitat: **Rajasthan** (Keoladeo) 2× 1996, 2010
- Nilgiri "cloud goats" (tahr): **Eravikulam** 1× 2005

Park	Animal
Kaziranga	One-horned rhino
Gir	Asiatic lion
Sundarbans	Tiger
Periyar	Elephant
Dachigam	Hangul
Eravikulam	Nilgiri tahr
Keibul Lamjao	Sangai deer
Keoladeo	Birds
Dudhwa	Swamp deer, rhino

11.4 Marine Parks and Miscellany

- Marine park: **Gulf of Mannar** 1× 2015
- Oceanic (first marine) park: **Kutch** 2× 2013, 2016
- Rani Jhansi Marine NP (A&N): **1996** 1× 2017
- UNESCO site July 2016: **Khangchendzonga** 1× 2016
- National Biological Garden: **New Delhi** 1× 2014
- Project Elephant: **1992** (no option, *) 1× 2007
- Garden of Five Senses: near **Mehrauli** 1× 2009
- Over 30% under national parks: **Bhutan** 1× 2015

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PARK-STATE TRAPS

Pin Valley
Himachal, not J&K

↓

Bandipur
Karnataka, not Tamil Nadu

↓

Similipal
Odisha, not MP

↓

Indravati
Chhattisgarh (Bastar)

↓

Rajiv Gandhi NP
Nagarhole, Karnataka

TRAP ALERT

- **Keoladeo** = birds, not lions; **Periyar** = elephants, not hangul.
- First **marine** park = Gulf of **Kutch**.
- Most national parks = **MP**; most sanctuaries = **A&N**.

UP LINK

Dudhwa National Park (Lakhimpur Kheri) is UP's only national park: swamp deer (barasingha), tiger, and **rhino rehabilitation** since 1984. UP's other big reserves are **Pilibhit, Amangarh, Ranipur** (tiger reserves) and **Katarniaghat** (sanctuary).

LAST-MINUTE 10

1. Corbett = first NP (1936), Ramganga
2. MP = most national parks
3. Kaziranga = rhino; Gir = lion
4. Dachigam = hangul; Eravikulam = Nilgiri tahr
5. **Dudhwa (UP)** = rhino rehabilitation, swamp deer
6. Keoladeo = birds, Siberian crane (Bharatpur)
7. Pin Valley = Himachal; Bandipur = Karnataka
8. Similipal = Odisha; Indravati = Chhattisgarh
9. Keibul Lamjao = Loktak, floating
10. Gulf of Kutch = first marine park

Quantum IAS & PCS

Chapter 12

Biosphere Reserves, Wildlife Sanctuaries and Project Tiger

Built from 47 UPPSC questions (1994–2025). Focus: biosphere reserve-state matching, UNESCO MAB, sanctuaries, Project Tiger, tiger reserve-state matching.

ASKED MOST · REVISE FIRST

- Biosphere reserves: **Nokrek-Meghalaya (Garo Hills), Manas-Assam, Similipal-Odisha, Agasthyamalai-Kerala, Dihang-Dibang-Arunachal, Pachmarhi-MP, Nanda Devi-Uttarakhand, Cold Desert-Himachal** 11× 2007, 2010, 2013, 2013, 2014, 2019, 2021, 2022, 2024, 2025, one-liner
- Tiger reserves: **Bandipur-Karnataka, Melghat-Maharashtra, Buxa-West Bengal, Pakke and Namdapha-Arunachal, Dampa-Mizoram, Sariska-Rajasthan; Amangarh is UP** 7× 2003, 2006, 2008, 2010, 2012, 2012, 2021
- **Project Tiger** = **1973**; Tiger Day **29 July** 4× 1994, 2002, 2014, 2018
- **Periyar** (Kerala) = wild elephants 2× 2011, 2015

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

12.1 Biosphere Reserves

- Preserve **genetic diversity** 1× 2004
- Correct: **biosphere-Edward Suess** (coined "biosphere") 1× 2013
- Not in UNESCO network (then): **Rann of Kutch** 1× 2009
- Largest UNESCO-listed: **Gulf of Mannar** 1× 2013
- **Nilgiri-1986, Nanda Devi-1988, Sundarbans-1989, Kanchenjunga-2000** 1× 2012
- **Nokrek-Meghalaya, Manas-Assam, Dihang-Dibang-Arunachal, Agasthyamalai-Kerala** 1× 2013
- **Similipal-Odisha, Nokrek-Meghalaya, Agasthyamalai-Kerala, Manas-Assam** 2× 2019, 2024
- Correct: **Nokrek-Meghalaya**; wrong: **Nokrek-Manipur** 2× 2007, 2025
- **Similipal-Odisha, Dihang-Dibang-Arunachal, Nokrek-Meghalaya, Cold Desert-Himachal** 1× one-liner
- **Similipal-Odisha, Pachmarhi-MP, Nanda Devi-Uttarakhand, Nokrek-Meghalaya** 1× 2021
- Wrong: **Manas-Meghalaya** (Assam) 1× 2013
- **Manas-Assam, Sundarbans-West Bengal, Nanda Devi-UP (then), Pachmarhi-MP** 1× 2022
- Spread over Garo Hills: **Nokrek** 1× 2014
- 16th reserve, Cold Desert: **Himachal Pradesh** 1× 2010
- World Heritage site: **Nanda Devi** 2× 2005, 2008
- **Bhitarkanika**: **Odisha** (Ramsar; only on UNESCO tentative list) 1× 2008

Biosphere reserve	State
Nilgiri (first, 1986)	TN, Kerala, Karnataka
Nanda Devi	Uttarakhand
Nokrek	Meghalaya
Gulf of Mannar	Tamil Nadu
Sundarbans	West Bengal
Manas	Assam
Similipal	Odisha
Dihang-Dibang	Arunachal Pradesh
Pachmarhi, Panna	Madhya Pradesh
Agasthyamalai	Kerala, TN
Kachchh (largest)	Gujarat
Cold Desert	Himachal Pradesh

12.2 Sanctuaries

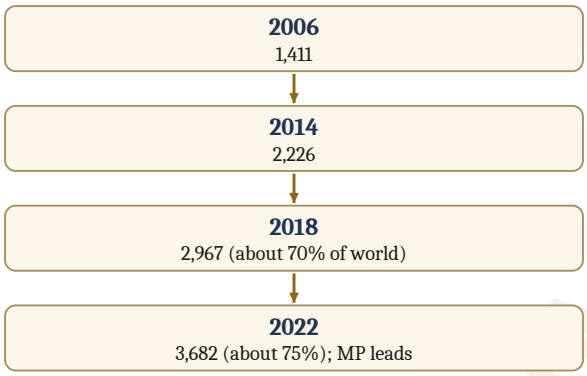
- **Periyar**: **wild elephants**; in **Kerala** 2× 2011, 2015
- **Chinnar** sanctuary: **Kerala** 1× 2005
- Wild ass sanctuary: **Gujarat** (Little Rann) 1× 2010
- **Kaziranga**: **rhinoceros** 1× 2012
- **Garampani-Assam, Namdapha-Arunachal, Pakhal-Andhra, Sariska-Rajasthan** 1× 2003
- **Namdapha-Arunachal, Bandipur-Karnataka, Periyar-Kerala, Lamjao-Manipur** 1× 2015
- **Dachigam-J&K, Keoladeo-Rajasthan, Kanha-MP, Periyar-Kerala** 1× 2014
- Wrong: **Similipal-MP** (Odisha) 1× 2009
- Tamil Nadu bird sanctuary: **Karikili** 1× 2008
- **National Chambal Sanctuary** is not in **Haryana** (it is MP, Rajasthan, UP) 1× 2020
- UP sanctuary of 1975: **Katarniaghat** (dates differ, *) 1× 2022
- **Shuklaphanta**: **Nepal** 1× 2016
- Largest botanical garden: **Kew** (England) 1× 2015

12.3 Project Tiger

- Aim: **save the Indian tiger** from extinction 1× 1994
- Started: **1973** 2× 2002, 2018
- International Tiger Day: **29 July** 1× 2014
- World Tiger Summit 2010: **St. Petersburg** 1× 2011

- Tiger census 2018: India has **70%** of world tigers; about **30%** outside reserves (both) 1× 2020
- Largest tiger habitat: **Andhra Pradesh** (Nagarjunsagar-Srisailam) 1× 2011
- **Bandipur-Karnataka, Melghat-Maharashtra, Buxa-West Bengal, Pakke-Arunachal** 1× 2008
- **Bandipur-Karnataka, Manas-Assam, Namdapha-Arunachal, Similipal-Odisha** 1× 2006
- Mizoram's tiger reserve: **Dampa** 1× 2012
- Wrong: **Buxa-Bihar** (West Bengal); Buxa = **West Bengal** 2× 2010, 2003
- Lies in two States: **Pench** (MP, Maharashtra) 1× 2008
- **Sariska**: **Rajasthan** 1× 2012
- Not in MP: **Amangarh** (UP) 1× 2021
- Newly declared (2015): **Rajaji** 1× 2015
- First to use drones: **Bandipur** 1× 2014

TIGER CENSUS (CENTRAL ESTIMATE)



TRAP ALERT

- **Buxa** = West Bengal; **Dampa** = Mizoram; **Amangarh** = UP.
- **Nokrek** = Meghalaya; **Manas** = Assam.
- **Chinnar** = Kerala, not Kashmir.

UP LINK

UP's tiger reserves: **Dudhwa, Pilibhit** (won TX2 award 2020 for doubling tigers), **Amangarh** (Bijnor) and **Ranipur** (Chitrakoot, 2022). The **National Chambal Sanctuary** covers UP, MP and Rajasthan; **Katarniaghat** guards gharials.

LAST-MINUTE 10

1. Nilgiri = first biosphere reserve (1986)
2. Nokrek = Meghalaya (Garo Hills); Manas = Assam
3. Cold Desert = Himachal; Dihang-Dibang = Arunachal
4. Gulf of Mannar = largest UNESCO-listed
5. Project Tiger **1973**; Tiger Day **29 July**
6. 2022 census **3,682**, ~75% of world
7. Nagarjunsagar-Srisailam = largest reserve
8. Buxa = WB; Dampa = Mizoram; Pakke = Arunachal
9. Pench = MP + Maharashtra; Amangarh = UP
10. Chambal sanctuary: MP, Rajasthan, UP

Chapter 13

Non-Conventional Energy

Built from 23 UPPSC questions (2000–2025). Focus: solar energy, India's energy mix, solar firsts, ISA, biofuels, hydrogen.

ASKED MOST · REVISE FIRST

- **Solar** = most eco-friendly, no air pollution, greatest renewable potential in India 5× 2004, 2006, 2011, 2015, 2018
- **Hydrogen** = least polluting fuel 4× 2010, 2013, 2015, 2017
- Solar firsts: **Cochin** airport; **Diu** first 100% solar UT 3× 2017, 2017, 2019
- **Jatropha** = biodiesel crop 1× 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

13.1 Sources

- Solar radiation's main role: **water cycle** 2× 2000, 2004
- Ultimate source of fossil-fuel energy: **Sun** 1× 2016
- Non-renewable: **petroleum** 1× 2021
- **Minerals-exhaustible, solar-inexhaustible, biogas-methane and CO2, deforestation-rainfall** 1× 2015
- Not biomass energy: **nuclear** 1× 2022
- Used most in organic form: **solar energy** (via plants) 1× 2004

13.2 India's Mix and Solar

- Second source after fossil fuels: **solar** 1× 2021
- Greatest renewable potential: **solar** 1× 2011
- Best for sustainable power (key): **hydro-electricity** 1× 2012
- Most eco-friendly: **solar cell** 3× 2004, 2006, 2015
- No air pollution: **solar** 1× 2018
- First fully solar airport: **Kochi** 1× 2017
- First 100% solar UT: **Diu** 2× 2017, 2019

Installed power (2025)	Share
Fossil	~48-50%
Solar	~26%
Hydro	~11%
Wind	~11%
Nuclear	~1.7%
Target 2030	500 GW non-fossil

13.3 International

- Decade of Sustainable Energy for All: **United Nations** 1× 2017
- ISA: World Solar Bank and OSOWOG under it (true); 38 members (false; over 100) (only 2) 1× 2025
- Largest offshore wind farm (2024): disputed (*) (UK's Hornsea 2 is largest) 1× 2023

13.4 Biofuels and Others

- Not correct: biofuel is **cost-effective** 1× 2014
- Biodiesel crop: **jatropha** 1× 2015
- Green fuel source: karanja/ethanol (*) 1× 2010
- Least polluting fuel: **hydrogen** 4× 2010, 2013, 2015, 2017
- First 100% organic state: **Sikkim** 1× 2019
- Nuclear projects must study **all** listed effects 1× 2013
- No geothermal sources: **Ganga delta** 1× 2016

INDIA'S CLEAN-FUEL MISSIONS

E20 ethanol
20% blend nationwide (2025)
↓
SATAT / GOBARdhan
Compressed biogas
↓
Green Hydrogen Mission (2023)
5 Mt/year by 2030
↓
PM Surya Ghar
Rooftop solar
↓
PM-KUSUM
Solar pumps for farmers

TRAP ALERT

- **Diu** = first solar UT; **Cochin** = first solar airport.
- **ISA** has 100+ members; HQ Gurugram.
- **Hydrogen** burns to water; least polluting.

UP LINK

UP's solar push: **Bundelkhand** solar parks (Jhansi, Lalitpur, Jalaun), **Ayodhya** as a model solar city, and UP Solar Policy 2022 targeting 22 GW; **Rampura** (2010) is the first solar village.

LAST-MINUTE 10

1. Sun = ultimate energy source
2. Solar = most eco-friendly, top potential
3. Solar is 2nd after fossil fuels
4. Kochi = first solar airport; Diu = first solar UT
5. Hydrogen = least polluting
6. Jatropha, karanja = biodiesel
7. Sikkim = first organic state
8. ISA: Gurugram; World Solar Bank, OSOWOG
9. Ganga delta: no geothermal
10. Nuclear energy is not biomass

Chapter 14

Pollution: Air, Acid Rain, Water, Sound and Soil

Built from 75 UPPSC questions (2000–2025). Focus: biodegradable pollutants, primary and secondary pollutants, CO, lichens, acid rain, arsenic and fluoride, BOD, decibels, Bhopal.

ASKED MOST · REVISE FIRST

- **Bhopal gas**: **methyl isocyanate** (MIC), 1984 6× 2001, 2002, 2008, 2014, 2017, 2022
- **Lichens** = bio-indicators of air pollution 7× 2010, 2012, 2013, 2015, 2019, 2021, 2025
- **Acid rain**: **SO2 and nitrogen oxides**; Norway and Canada hit hardest 10× 2000, 2001, 2001, 2002, 2003, 2007, 2010, 2011, 2013, 2024
- **Carbon monoxide** binds haemoglobin; most dangerous 4× 2006, 2013, 2014, 2016
- **Fly ash** comes from **burning coal** 3× 2004, 2004, 2021
- **BOD** measures water pollution; high BOD = sewage 4× 2007, 2011, 2021, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Basic Ideas

- Human-made pollutants: **anthropogenic**; human-modified environment: anthropogenic 2× 2005, 2019
- Biodegradable: **sewage**; **rubber** (of options) 3× 2014, 2016, 2014
- **DDT**: **non-biodegradable** 1× 2022
- Not biodegradable: **pesticide** 1× 2023
- Polythene lasts because it is a **polymer** 1× 2007
- Takes longest to decay: **plastic bag** 1× 2008

14.2 Air Pollution

- Burning coal, petrol, diesel: **air pollution** 1× 2011
- Fossil-fuel air pollutant: **SO2** 1× 2011
- Not a water pollutant: **SO2** 1× 2011
- Not secondary: **SO2** (primary) 1× 2014
- Secondary pollutant: **smog** 1× 2018
- Eye irritant in smoke: **PAN** (peroxyacetyl nitrate) 1× 2018
- Photochemical smog always has **ozone** 1× 2022
- Common atmospheric pollutant: **CO** 1× 2006
- Affects blood, can kill: **CO**; binds haemoglobin faster than O2; most dangerous 3× 2013, 2016, 2014
- Fly ash: **burning coal** 3× 2004, 2004, 2021
- Lichens: good indicators (true); grow well in pollution (false) (only 1) 1× 2025
- Air-pollution indicator: **lichen**; bio-indicator 6× 2010, 2013, 2021, 2012, 2015, 2019
- Most polluted city (then): **Ankleshwar** 1× 2009
- Greenpeace 2020 most polluted: **Jharia** 1× 2019
- Cleans polluted air: **rain** 1× 2013
- Delhi winter pollution peaks (true); more vehicle combustion in winter (false) 1× 2004
- Cigarette smoke: **CO and benzene** (per key) 1× 2015
- Not an air-pollution control device: **cyclone divider** 1× 2013
- Bio-decomposer: free to farmers; made of fungi (both) 1× 2025
- **NCAP**: launched **2019** 1× 2020
- **PRANA** portal: monitors **NCAP** 1× 2023

Primary	Secondary
SO2, CO, NOx	Ground-level ozone
Hydrocarbons	PAN
Particulates, lead	Photochemical smog, acid rain

14.3 Acid Rain

- CO2, SO2, NO2 dissolve in rain; polluted air (R explains A) 1× 2024
- Acid rain: **nitrogen oxides and SO2**; clouds react with SO2 5× 2001, 2007, 2010, 2013, 2001
- Main gas: **SO2** 2× 2002, 2003
- Suffer acid rain: **Canada and Norway** 1× 2011
- Most acid rain: **Norway** 1× 2000

14.4 Water Pollution

- Non-biotic groundwater pollutant: **arsenic** 1× 2012
- Arsenic highest in **Kolkata**; **West Bengal**; natural arsenic in **Bangladesh** 3× 2005, 2015, 2022
- **Knock-knee**: **fluoride** 1× 2016
- Ground Water Quality Report 2024: **Central Ground Water Board** (only 3) 1× 2025
- Worst chemical effluents: **leather** industry 1× 2005
- Not air pollution: **eutrophication** 1× 2013
- Eutrophication: nutrient enrichment; depletes oxygen (both) 1× 2019
- High BOD: **sewage** contamination 1× 2007
- River pollution measured by **dissolved oxygen** 2× 2011, 2021
- BOD: **pollution assay** of water 1× 2023
- Ganga BOD highest: **Kanpur to Allahabad** 1× 2008
- "Biological desert": **Damodar** 1× 2012
- UP's "biological disaster" river: **Yamuna** 1× 2018
- Largest solid waste per head: **Mumbai** 1× 2010

14.5 Noise and Soil

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- Noise unit: **decibel** 1× 2017
- Loudest: **jet flight** 1× 2003
- Rustling leaves: **20 dB** 1× 2018
- **Green muffler**: **noise pollution** (tree belts) 2× 2014, 2017
- Overuse of fertilisers causes **all** listed harms 1× 2016
- Soil pollution worse; fertilisers enter the food chain (R explains A) 1× 2010
- Crop rotation helps in **all** listed ways 1× 2016

14.6 Pollutants and Diseases

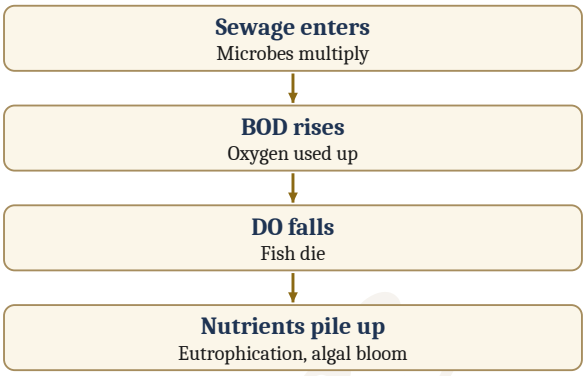
- Bhopal (1984): **methyl isocyanate** 6× 2001, 2002, 2008, 2014, 2017, 2022
- Wrong: **fluoride-indigestion** (fluorosis) 1× 2017
- Most toxic metal in exhaust: **lead** 2× 2006, 2009
- **Asbestos-lung, lead-brain, mercury-stomach, CO-bloodstream** 1× 2012
- **CO-liver and kidney, NOx-cancer, soil particles-breathing, lead-nervous system** (per key) 1× 2008
- **Acid rain-SO2, smog-unsaturated hydrocarbons, haemoglobin-CO, ozone depletion-CFCs** 1× 2019
- Radiation: **heredity change, blood circulation, cancer** (1, 3, 4) 1× 2009
- RSPM size: **less than 5 µm** (per key; PM10 and PM2.5 today) 1× 2017

Pollutant	Disease
Mercury	Minamata
Cadmium	Itai-itai
Nitrate	Blue baby
Fluoride	Fluorosis, knock-knee
Arsenic	Black-foot, skin cancer
Asbestos	Asbestosis
Silica	Silicosis
CO	Blocks haemoglobin

14.7 Vehicle Norms and Miscellany

- Euro norms cover **all** listed gases 1× 2013
- Euro-II diesel sulphur: **0.05% or less** 1× 2013
- Cold wave of January 2012: **La Nina** 1× 2010
- Mustard seed adulterant: **argemone** seed 1× 2010
- **Incinerators**: **burning waste** 1× 2018

READING WATER POLLUTION



TRAP ALERT

- **SO2 is primary**; smog, ozone, PAN are secondary.
- High **BOD**, low **DO** = polluted.
- **Eutrophication** is water pollution.
- Lichens **vanish** where air is dirty.

UP LINK

UP links: **Kanpur** tanneries push Ganga BOD highest between **Kanpur and Prayagraj**; the **Yamuna** is called a "biological disaster"; UP cities (**Ghaziabad, Noida, Lucknow**) are NCAP non-attainment cities; Sonbhadra's **Singrauli** belt has fly-ash and mercury problems.

LAST-MINUTE 10

1. Bhopal 1984 = **MIC**
2. Lichens = air-pollution indicator
3. Acid rain = SO2 + NOx; Norway, Canada
4. CO binds haemoglobin
5. Fly ash = coal; SO2 = primary pollutant
6. PAN = eye irritant; smog has ozone
7. Arsenic = West Bengal, Bangladesh; fluoride = knock-knee
8. BOD high = sewage; Damodar = biological desert
9. Noise in **decibels**; leaves 20 dB; green muffler
10. NCAP **2019**; PRANA portal

Chapter 15

Water Conservation and River Cleaning

Built from 13 UPPSC questions (2002–2023). Focus: World Water Day, Water Act and Cess, chlorine, lake conservation, Ganga and Yamuna action plans.

ASKED MOST · REVISE FIRST

- **Ganga Action Plan** and **Central Ganga Authority**: **1985**; aim: pollution-free water **2×** 2014, 2022
- **Water Act 1974**; Water Cess **1977** **2×** 2013, 2023
- **World Water Day**: **22 March** **1×** 2011

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

15.1 Water as a Resource

- World Water Day: **22 March** **1×** 2011
- Not a biotic resource: **freshwater** **1×** 2016
- CGWA under the **Environment Protection Act**; India has the largest groundwater-irrigated area (2 and 3) **1×** 2005

15.2 Laws

- Water (Prevention and Control of Pollution) Act: **1974** **1×** 2023
- Water Cess enforced: **1977** **1×** 2013

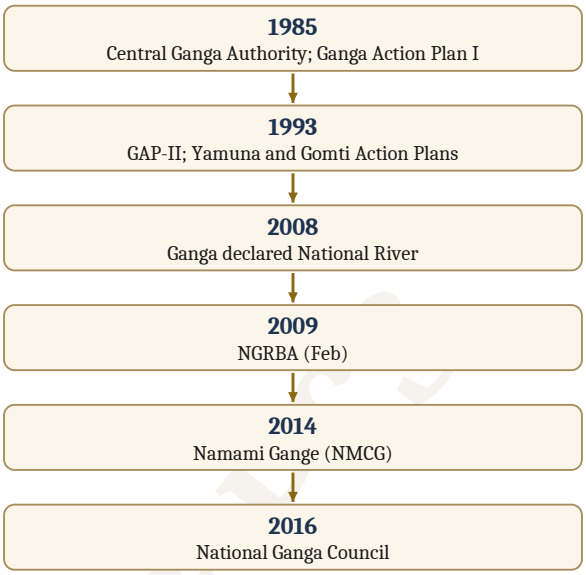
15.3 Treatment

- Drinking-water purification gas: **chlorine** **1×** 2013
- Disinfectants: **all** listed (chlorine, ozone, UV) **1×** 2016

15.4 Lakes and Rivers

- Urban lakes under NLCP: **Bhoj, Sukhna, Pichola** (not Chilika) **1×** 2002
- Added to NLCP (2010): **Bhimtal** **1×** 2010
- Central Ganga Authority: **1985** **1×** 2014
- Ganga Action Plan aim: **pollution-free water only** **1×** 2022
- **NGRBA**: **February 2009** **1×** 2010
- Yamuna Action Plan: **1993** **1×** 2016

CLEANING THE GANGA



Day	Date
World Wetlands Day	2 February
World Water Day	22 March
World Environment Day	5 June
Ozone Day	16 September

TRAP ALERT

- **Water Act 1974**; **Cess Act 1977**; Air Act 1981.
- **NGRBA** = 2009; **Namami Gange** = 2014.
- **Chilika** is not an urban lake in the NLCP list.

UP LINK

The Ganga and its tributaries (**Yamuna, Gomti, Ramganga, Kali**) flow through UP; Kanpur's tanneries and Varanasi's ghats are the core of GAP and **Namami Gange**. The **Gomti Action Plan** (1993) covers Lucknow.

LAST-MINUTE 10

1. World Water Day **22 March**
2. Water Act **1974**; Cess **1977**
3. CGWA under **EPA 1986**
4. Chlorine purifies drinking water
5. Bhoj, Sukhna, Pichola = NLCP urban lakes
6. Ganga Action Plan & CGA **1985**
7. Yamuna Action Plan **1993**
8. NGRBA **Feb 2009**
9. Namami Gange **2014**
10. Freshwater = abiotic resource

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Chapter 16

Organisations, Awards, Days and Miscellaneous

Built from 54 UPPSC questions (2000–2025). Focus: institute-city matching, UN body headquarters, environment awards, important days, Mission LiFE, environmental movements.

ASKED MOST · REVISE FIRST

- ■ **NEERI**: **Nagpur** (1958) 6× 2004, 2005, 2011, 2012, 2014, 2017
- ■ **World Environment Day**: **5 June** 5× 2004, 2011, 2012, 2014, 2025
- ■ Day matching: **Wetlands 2 Feb, Wildlife 3 March, Water 22 March, Health 7 April, Heritage 18 April, Biodiversity 22 May, No Tobacco 31 May, Population 11 July, Tiger 29 July, Ozone 16 Sept, Soil 5 Dec** 7× 2008, 2017, 2018, 2022, 2023, 2023, 2025
- ■ HQs: **UNEP-Nairobi, WMO-Geneva, IMO-London, Greenpeace-Amsterdam** 4× 2005, 2010, 2010, 2016
- • **CAZRI**: **Jodhpur** 2× 2008, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

16.1 Organisations

- ■ NEERI: **Nagpur**; set up **1958** 6× 2004, 2005, 2011, 2012, 2014, 2017
- • **WMO**: **Geneva** 1× 2010
- • **UNEP**: **Nairobi** 1× 2010
- • **Greenpeace International**: **Amsterdam** 1× 2005
- • **IMO**: **London** 1× 2016
- • **Nutrition-Hyderabad, Wildlife Institute-Dehradun, Ayurveda-Jaipur, Naturopathy-Pune** 1× 2004
- • Wrong: **IIFM-Kolkata** (it is **Bhopal**) 1× 2012
- • ICFRE pairs: Arid-Jodhpur, Rain-Jorhat, Tropical-Jabalpur right; Himalayan is **Shimla**, not Dehradun (3 correct) 1× 2023
- • **CAZRI**: **Jodhpur** 2× 2008, 2010
- • Indian Institute of Ecology and Environment: **New Delhi** 2× 2014, 2016
- • ENVIS centre on population and environment: **IIPS Mumbai** 1× 2017
- • India's working Antarctic stations: **Bharati and Maitri** 1× 2021
- • First National Centre for Marine Biodiversity: **Jamnagar** 2× 2018, 2023

Body	HQ
UNEP	Nairobi
WMO, IPCC	Geneva
IMO	London
IUCN, WWF	Gland
Greenpeace	Amsterdam
UNFCCC	Bonn
ISA	Gurugram

16.2 Awards

- • **Indira Gandhi Paryavaran Puraskar**: contribution to the **environment** 1× 2004
- • **Rajiv Gandhi Environment Award**: **clean technology** 1× 2006
- • **Borlaug Award**: **agriculture**; Borlaug's Nobel: **Peace** (1970) 2× 2001, 2018
- • **Tyler Prize**: **environmental protection** 1× 2004
- • World Biodiversity Day: **22 May** 1× 2018
- • **Global 500**: **environment conservation** 1× 2019
- • **Champions of the Earth** 2018: **Narendra Modi** (with Macron) 1× 2017

16.3 Days

- • World Habitat Day: first Monday of October (*) 1× 2000
- • Anti-Tobacco Day: **31 May**; "5 May" is wrong 2× 2008, 2017
- • **19 November**: **World Toilet Day** 1× 2015
- ■ World Environment Day: **5 June** 4× 2004, 2011, 2012, 2014
- • WED 2018 theme: **Beat Plastic Pollution** (India host) 1× 2017
- • World Wetlands Day: **2 February** 1× 2023
- • Wrong: **World Wildlife Day-22 March** (it is **3 March**) 1× 2025
- • **Population-11 July, Wildlife-3 March, Ozone-16 September, Health-7 April** 1× 2023
- • **18 April-Heritage, 22 May-Biodiversity, 29 July-Tiger, 5 December-Soil** 1× 2022
- • National Dolphin Day: **5 October** 1× 2022
- • Cleanliness Day on the last Saturday monthly: **Sierra Leone** 2× 2002, 2003

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ENVIRONMENT CALENDAR



16.4 Miscellany

- **Mission LiFE**: launched June 2022 (true); proposed at COP25 (false; **COP26**) (only 1) 1× 2025
- Eco-friendly lifestyle initiative: **Mission LiFE** 1× 2023
- India's World Heritage sites by Aug 2010: **28** 1× 2009
- Newly listed (2002): **Mahabodhi Temple** 1× 2002
- Wrong: **Silent Valley-Baba Amte** (Narmada) 1× 2009
- **Appiko**: Karnataka (key also Kerala) 1× 2025
- "Rally for the Valley": **resettlement** (Narmada) 1× 2003
- **Mitti Bachao**: **Hoshangabad**, MP 1× 2020
- Land degradation neutrality by **2030** 1× 2019
- Anthropogenic biome: **cropland** 1× 2018
- Collects and releases water: **watershed** 1× 2022
- Not eco-friendly: **eucalyptus** 1× 2022
- Dust-pollution tree: **Sita Ashok** 1× 2007
- Evergreen fruit plant: **loquat** 1× 2008
- Mega-diverse country: **Australia** 1× 2015
- **SODAR**: **Kaiga and Kalpakkam** 1× 2004
- **Skin cancer-UV, noise-decibel, global warming-CO2, ozone hole-CFC** 1× 2015
- "Father of Economic Ecology" (UNEP): **M.S. Swaminathan** 1× 2007
- Peninsular glaciation: **Pleistocene** 1× 2010
- India's largest charkha: **Noida**, made of **plastic waste** (both) 1× 2019

Movement	Leader / place
Chipko (1973)	Bahuguna, Bhatt; Chamoli (then UP)
Appiko (1983)	Panduranga Hegde; Karnataka
Silent Valley	KSSP, Sugathakumari; Kerala
Narmada Bachao	Medha Patkar, Baba Amte
Bishnoi (1730)	Amrita Devi; Khejarli
Mitti Bachao (1977)	Hoshangabad, MP

TRAP ALERT

- **UNEP** = Nairobi, not Geneva.
- **IIFM** = Bhopal; **Himalayan FRI** = Shimla.
- **Wildlife Day 3 March**; Water Day 22 March.
- **Borlaug** won the Nobel **Peace** Prize.

UP LINK

UP links: **Chipko** began in Chamoli when it was part of UP; ICFRE's **Forest Research Centre for Eco-Rehabilitation** is at **Prayagraj**; India's largest **charkha** of plastic waste stands in **Noida**.

LAST-MINUTE 10

1. NEERI = **Nagpur** (1958); CAZRI = **Jodhpur**
2. UNEP = Nairobi; WMO = Geneva; IMO = London
3. Greenpeace = Amsterdam; IIFM = Bhopal
4. NCMB = **Jamnagar**; Antarctic: Bharati, Maitri
5. Environment Day **5 June**; Ozone Day **16 Sept**
6. Wildlife Day **3 March**; Tobacco Day **31 May**
7. Dolphin Day **5 Oct**; Toilet Day **19 Nov**
8. Tyler Prize = environment; Champions of the Earth 2018 = Modi
9. Mission LiFE = June 2022 (proposed COP26)
10. Swaminathan = Father of Economic Ecology